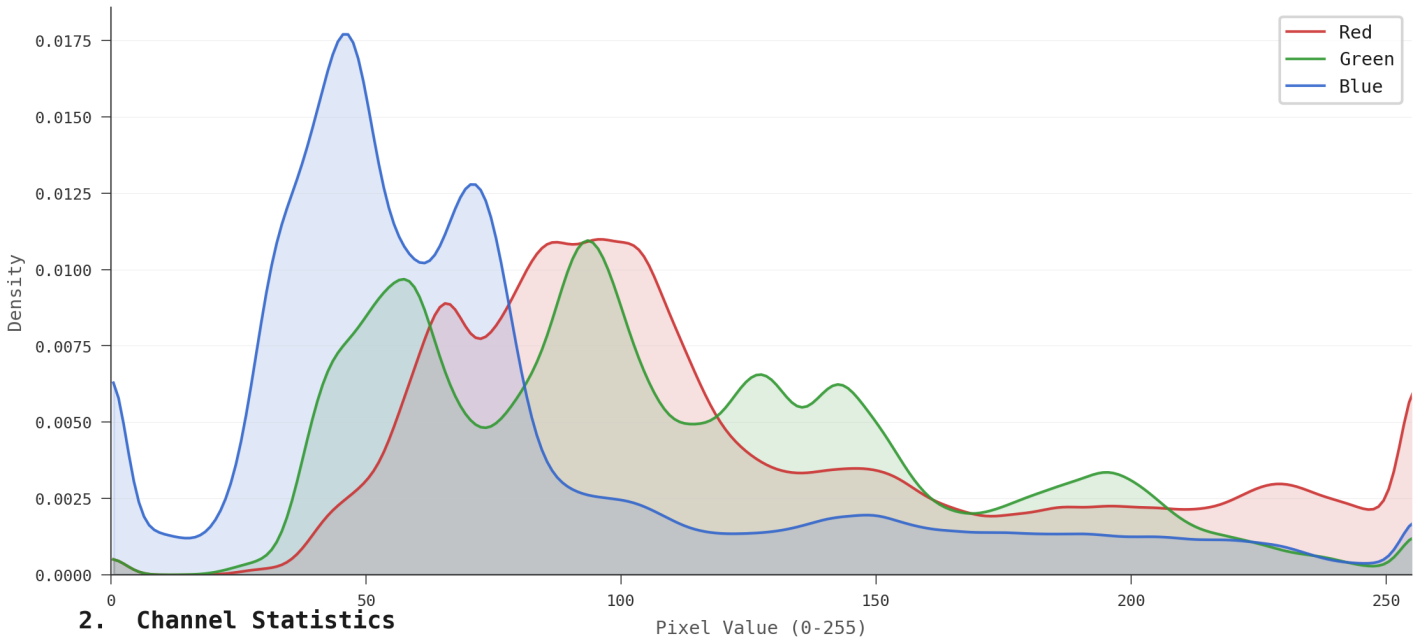
Reference Data Sheet: KODIM23

Subject: Macaw Pair on Perch

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	121.6621	57.4511	3300.6241	-0.3263	0.8485	0	255
Green	109.5960	50.8010	2580.7367	-0.3619	0.5988	0	255
Blue	75.7940	53.6149	2874.5523	1.5599	1.4627	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.6147	0.4231
G	0.6147	1.0000	0.6408
B	0.4231	0.6408	1.0000

Pairwise Pearson Correlation Coefficients

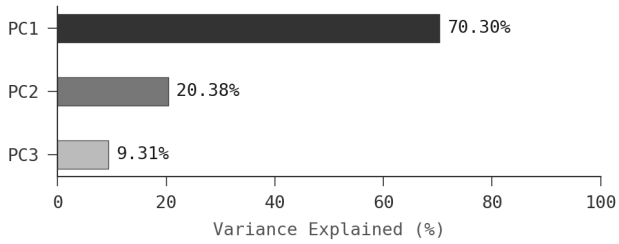
r(R-G)	=	**0.614747**
r(R-B)	=	**0.423120**
r(G-B)	=	**0.640769**
Avg |r|	=	**0.559545**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{3300.6325} & 1794.1847 & 1303.3100 \\
1794.1847 & \mathbf{2580.7432} & 1745.2565 \\
1303.3100 & 1745.2565 & \mathbf{2874.5596}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM23

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	**-0.6100**	-0.5732	-0.5472	6155.8496
PC2	-0.7170	0.1051	0.6891	1784.8463
PC3	0.3375	-0.8126	0.4751	815.2394

6. Derived Metrics

Condition Number

7.55

(l1/l3) PC1/PC2: 3.45 | PC2/PC3: 2.19

Blue Channel Independence

35.9%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Red

Loading: -0.6100 | Red dominant

Classification: Three-Dimensional (PC1 = 70.30%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.988969	0.984640	0.986804
Green	0.986138	0.980135	0.983137
Blue	0.986545	0.984012	0.985279

8. Average Local Variance (3x3)

Channel	Value
Red	76.9947
Green	75.5735
Blue	75.0538

9. Redundancy Profile

Classification: Three-Dimensional | Condition Number: 7.55 | PC1: 70.30%

Correlation: G–B pair leads at r = 0.640769 | R–G: 0.614747 | R–B: 0.423120

Blue Channel: 35.9% independent | 3D color; major blue variance

Spatial: Avg autocorrelation: 0.985073 | Avg local variance: 75.8740

Eigenvector: Red dominant | R (-0.6100) leads | G (-0.5732), B (-0.5472)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."